

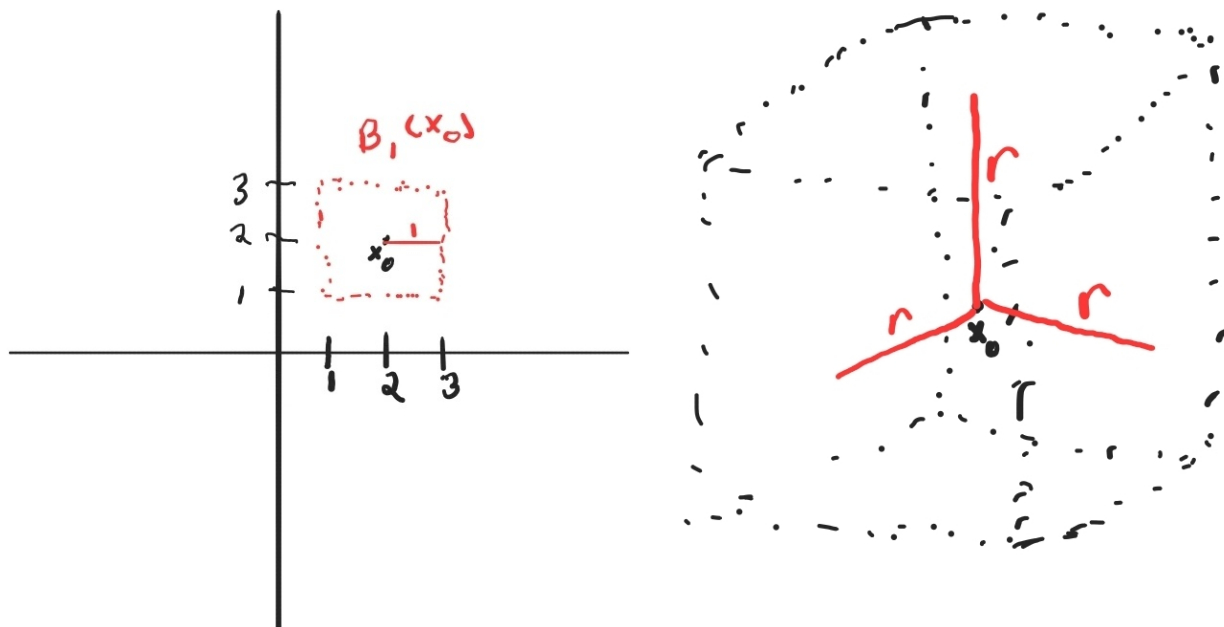
Homework 1

Connor Mooneyhan

January 22, 2026

Exercise 1. On $\mathbb{R}^2$, let $d : X \times X \rightarrow \mathbb{R}$ be the box metric. Sketch the ball $B_r(x_0)$ where x_0 is a point in $\mathbb{R}^2$ and $r > 0$ is a real number. Do the same for $\mathbb{R}^3$.

Proof. I left out the axes on the second one because it was already crowded enough. The one on the left is an open square with side lengths 2 (where I've chosen $r = 1$), and the one on the right is a general open cube with side lengths $2r$. Both have x_0 in the center.



□

Exercise 2. On $\mathbb{R}$, is $d(x, y) = \sqrt{|x - y|}$ a metric?

Proof. Positivity is guaranteed because we are taking principal square roots of nonnegative values $|x - y|$, and definiteness is guaranteed because

$$\sqrt{|x - y|} = 0 \implies |x - y| = 0 \implies x - y = 0 \implies x = y.$$

The definition is inherently symmetric since $|x - y| = |y - x|$. Now assume we have $x, y, z \in \mathbb{R}$.

Then $\sqrt{|x - z|} \leq \sqrt{|x - y| + |y - z|}$ by the triangle inequality for absolute value, and

$$\begin{aligned} \sqrt{|x - y| + |y - z|} &\leq \sqrt{|x - y|} + \sqrt{|y - z|} \\ \iff \left(\sqrt{|x - y| + |y - z|}\right)^2 &\leq \left(\sqrt{|x - y|} + \sqrt{|y - z|}\right)^2 \\ \iff |x - y| + |y - z| &\leq |x - y| + 2\sqrt{|x - y||y - z|} + |y - z| \\ \iff 0 &\leq 2\sqrt{|x - y||y - z|}. \end{aligned}$$

Since the last line is always true, the first line is also true, which implies that

$$d(x, z) \leq d(x, y) + d(y, z).$$

Thus d is a metric. □

Exercise 3. Let X be the set of all ordered n -tuples of zeroes and ones. In other words, $X = \{0, 1\}^n$. Define

$$d(x, y) = \text{number of places where } x \text{ and } y \text{ have different entries.}$$

Show that this is a metric on X . This metric is known as the Hamming distance and plays an important role in switching and automata theory and coding.

Proof. The phrase “the number of” takes care of positivity. If $d(x, y) = 0$, then x and y differ in zero entries, so they are equal, and thus d is definite. The definition of d is also inherently symmetric since the number of different entries does not change if we switch the order in which we compare the entries of x and y .

Now, if we have $x, y, z \in X$, then suppose the set of entries in which x is different from y is disjoint from the set of entries in which y and z are different. Then $d(x, z) = d(x, y) + d(y, z)$. If the aforementioned sets overlap, then $d(x, z) \leq d(x, y) + d(y, z)$. In either case, we have $d(x, z)$ is at most $d(x, y) + d(y, z)$, so the triangle inequality holds, and d is indeed a metric. □

Exercise 4. Let (X, d) be a metric space. For any non-empty subset $A \subset X$, define the *diameter* of A as

$$\delta(A) = \sup_{x, y \in A} d(x, y).$$

A set A is called bounded if $\delta(A) < \infty$. Show that if $A \subset B$ then $\delta(A) \leq \delta(B)$. Also, give an example of a metric space and two sets $A \subsetneq B$ with $\delta(A) = \delta(B)$.

Proof. If $A \subset B$, then $\{d(x, y)\}_{x, y \in A} \subset \{d(x, y)\}_{x, y \in B}$, so since $\delta(B)$ is an upper bound of the latter set, it is an upper bound of the former, and thus $\delta(A) \leq \delta(B)$ since $\delta(A)$ is the least such upper bound.

Now consider the trivial/discrete metric on $\mathbb{R}$, and let $A = (-1, 1)$ and $B = (-2, 2)$. Then we have $A \subsetneq B$ and $\delta(A) = \delta(B) = 1$. □

Exercise 5. On any metric space X , prove that $\delta(A) = 0$ if and only if A consists of a single point.

Proof. If $\delta(A) = 0$, then the positivity of d tells us that we must have $d(x, y) = 0$ for all $x, y \in A$, and definiteness guarantees that any such pair satisfies $x = y$, so A consists of one point. On the other hand, if A consists of a single point x , then indeed

$$\delta(A) = \sup_{x, y \in A} d(x, y) = \sup\{d(x, x)\} = \sup\{0\} = 0.$$

□

Exercise 6. Let (X, d_X) , (Y, d_Y) be two metric spaces. On the Cartesian product $X \times Y$, define, for $\mathbf{x}_1 = (x_1, y_1)$ and $\mathbf{x}_2 = (x_2, y_2)$,

$$d(\mathbf{x}_1, \mathbf{x}_2) = \max(d_X(x_1, x_2), d_Y(y_1, y_2)).$$

Show that d is a metric on $X \times Y$.

Proof. Since $d(\mathbf{x}_1, \mathbf{x}_2) \geq d_X(x_1, x_2) \geq 0$, we have positivity of d . Similarly, if $d(\mathbf{x}_1, \mathbf{x}_2) = 0$, then both $d_X(x_1, x_2)$ and $d_Y(y_1, y_2)$ must be 0, which implies that $x_1 = x_2$ and $y_1 = y_2$, so $\mathbf{x}_1 = \mathbf{x}_2$. By the symmetry of d_X and d_Y ,

$$d(\mathbf{x}_1, \mathbf{x}_2) = \max(d_X(x_1, x_2), d_Y(y_1, y_2)) = \max(d_X(x_2, x_1), d_Y(y_2, y_1)) = d(\mathbf{x}_2, \mathbf{x}_1).$$

I'll change notations for the triangle inequality. Let $x = (x_1, x_2)$, $y = (y_1, y_2)$, and $z = (z_1, z_2)$. Then

$$\begin{aligned} d(x, z) &= \max(d_X(x_1, z_1), d_Y(x_2, z_2)) \\ &\leq \max(d_X(x_1, y_1) + d_X(y_1, z_1), d_Y(x_2, y_2) + d_Y(y_2, z_2)) \\ &\dots \\ &\leq \max(d_X(x_1, y_1), d_Y(x_2, y_2)) + \max(d_X(y_1, z_1), d_Y(y_2, z_2)) \\ &= d(x, y) + d(y, z) \end{aligned}$$

where the third line follows from the positivity of d_X and d_Y . Thus d is a metric. □